

HEREDITY AND ENVIRONMENT

Genetics - the study of passing on of characteristics from one generation to the next.

Heredity - the passing on of characteristics from parents to their offspring.

Scientists use the study of genetics to show how variation in characteristics occurs in a particular species.

Variation occurs within a species due to:

(a) ***inheritable characteristics from the parent(s).***

e.g. face shape, eye colour, flower colour, etc.

(b) ***environmental factors.***

e.g. climate, food source or supply, water, etc.

Example

Children who live in areas with high lead contamination do not show the same increase in intellectual abilities as the average population.

Hence lead has been phased out of petrol. It was the main form of lead pollution in the air.

Inheritable characteristics depend on the method of reproduction used by the organism.

- ***Asexual*** - one parent only.

The genetic information of the offspring is identical to the parent.

- ***Sexual*** - two parents.

The offspring has a mixture of information from both parents.